

Pradeep Murugesan

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Summary

AI Engineer specializing in real-time computer vision and GenAI systems, building low-latency GPU-accelerated inference pipelines and scalable microservice architectures. Experienced in deploying end-to-end AI solutions from model training and ONNX/TensorRT optimization to production-grade FastAPI services and RAG-based AI systems using LangChain and vector databases. Experienced with MLOps practices including Docker, Kubernetes, and CI/CD pipelines, with a focus on reliable backend engineering and cloud deployment.

Technical Skills

- **Computer Vision:** YOLO, YOLO-Pose, OpenCV, ONNX, OpenVINO, TensorRT, EfficientNet
- **Machine Learning:** PyTorch, NumPy, Pandas, Matplotlib
- **Generative AI:** LangChain, LangFlow, RAG Pipelines, Embeddings, Qdrant, Pinecone, Local Ollama, vLLM
- **Backend:** FastAPI, Spring Boot, SQLAlchemy, REST API Development, Microservices, RabbitMQ, Redis
- **MLOps / DevOps:** Docker, Kubernetes (AKS), Container Orchestration, Nginx, GitHub Actions, CI/CD, Cloud Deployment, Git, Linux, Postman, Swagger
- **Databases:** PostgreSQL, MySQL, Minio, Supabase
- **Languages:** Python, Java, SQL
- **AI Agents / Tools:** Claude, GitHub Copilot, Codex, Gemini CLI

Experience

AI Engineer | Global Tech Professionals, London Mar 2025 – Present
Remote

- Developed real-time computer vision systems using YOLO and YOLO-Pose, integrated with FastAPI for low-latency inference for real-time video processing applications.
- Built asynchronous video processing pipelines for model inference and request queuing, improving throughput for batch inference workloads.
- Optimized model performance using TensorRT conversion and batching techniques for faster GPU inference.
- Implemented and maintained REST APIs and backend services supporting AI workloads, enabling reliable integration with downstream systems.
- Developed a real-time chatbot integrated with the backend services to deliver instant, AI-driven responses to user queries.

Freelance Computer Vision Engineer Jul 2024 – Mar 2025
Computer Vision & Deep Learning | Remote

- Delivered 3+ computer vision solutions for object detection, pose estimation, and motion analysis using Python, OpenCV, and YOLO with GPU-accelerated inference.
- Improved model performance across projects through hyperparameter tuning, dataset augmentation, and transfer learning; deployed inference services using FastAPI and Docker.

Java Back End Developer Intern | SplendensLab IT Ventures, Salem Jan 2024 – Jun 2024
On-site

- Developed REST APIs using Spring Boot and implemented CRUD operations with PostgreSQL database integration.
- Built backend modules following MVC architecture and full development lifecycle practices.
- Validated APIs using Postman and documented endpoints with Swagger.

Projects

WoundCare-AI | Intelligent Wound Assessment & Monitoring System Mar 2026 – Present
github.com/solnae-tech | *Group Project – Backend & AI Engineer*
Stack: Flutter, FastAPI, RabbitMQ (CloudAMQP), PostgreSQL, Docker, GitHub Actions, OpenCV, YOLO (Segmentation), EfficientNet

- Contributed to a cross-functional team building a multi-service platform with patient, AI, notification, and chatbot components.
- Built YOLO instance segmentation and EfficientNet transfer learning pipelines for wound boundary detection and severity classification, integrated with RabbitMQ-based asynchronous processing.
- Designed and deployed FastAPI backend services with JWT-secured endpoints, containerized via Docker, and automated CI/CD pipelines with GitHub Actions.

ESTIMAX-AI | AI Blueprint Estimation System

Jan 2026 – Mar 2026

github.com/EstimaX-AI | *Group Project – Backend & AI Engineer*

Stack: Flutter, FastAPI, RabbitMQ (CloudAMQP), Supabase, Modal Workers, Render, Docker, GitHub Actions, OpenCV, YOLO

- Contributed to system design and development for a distributed AI estimation platform spanning mobile clients, FastAPI microservices, and cloud inference workers.
- Implemented YOLO and OpenCV pipelines for engineering symbol detection in blueprint drawings, coordinating asynchronous inference jobs through CloudAMQP.
- Automated CI/CD workflow with GitHub Actions and Docker to support reliable deployments across backend and AI services.

FAQ-Bot | RAG-Based Question Answering System

May 2026

github.com/Mpradeep-dev/FAQ-bot | *Individual Project*

Stack: Qdrant, Embeddings, Local Ollama, Redis

- Built a Retrieval-Augmented Generation (RAG) FAQ chatbot that grounds answers in a knowledge base using semantic embeddings and vector search.
- Designed the retrieval pipeline with document chunking, embedding generation, and similarity search over a Qdrant vector database for accurate context retrieval.
- Ran a local LLM with Ollama for fully offline, low-cost inference without external API dependencies.
- Engineered context-aware prompt construction to combine retrieved passages with user queries, reducing hallucinations and improving answer relevance.
- Implemented Redis-backed conversational memory retaining the last 5 messages to maintain context across multi-turn conversations.

AI Sports Trainer | Real-Time Coaching System (Gym)

Sep 2025

github.com/Mpradeep-dev/AI_Trainer | *Individual Project*

Stack: YOLO-Pose, OpenCV, Python

- Developed a real-time AI-based coaching system using YOLO-Pose for human pose estimation and landmark tracking.
- Analyzed body posture and movement patterns to provide instant visual feedback for gym exercises and workout form correction.
- Implemented frame-by-frame processing with OpenCV to achieve smooth real-time performance.
- Designed a modular pipeline for pose detection, feature extraction, and feedback generation.

FocusTrack | Real-Time Driver Drowsiness Detection

Jul 2023 – Feb 2024

github.com/Mpradeep-dev/drowsiness-detection | *Individual Project*

Stack: OpenCV, Dlib, Python

- Developed a real-time driver fatigue detection system using Dlib facial landmark detection and Eye Aspect Ratio (EAR) analysis.
- Implemented adaptive thresholding to improve detection robustness under varying lighting and user conditions.
- Achieved real-time alerting at approximately 30 FPS with continuous eye-state monitoring.
- Integrated an alert mechanism to notify users upon detecting prolonged eye closure or drowsiness patterns.

Education

M.Tech in Computer Science and Engineering

Expected 2027

Erode Sengunthar Engineering College, Perundurai CGPA: 7.9 / 10.0

Certifications

- Computer Vision 101 — Infosys Springboard
- Computer Vision with Python — Udemy
- Overview of APIs & Microservices — Infosys Springboard
- RESTful API Design — Infosys Springboard
- Python for Data Analysis — Udemy
- Mastering MySQL — Udemy
- GitHub Professional Certificate — GitHub

Achievements

- 1st Place, Codathon — Kongu Engineering College (2025)